THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 12	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$: 3+4+1+4=12.
2	**(a)** increases 100 times	Each unit of pH is a factor of 10.
3	**(c)** silver	Silver lies below hydrogen.
4	**(d)** 13	3n + 1 = 13.
5	**(b)** water	Splitting of water.
6	**(a)** receptor -> sensory neuron -> spinal cord -> motor neuron -> effector	Standard reflex pathway.
7	**(c)** Offspring produced by budding in Hydra	Asexual offspring are clones with minor copying errors.
8	**(b)** 1/2	Offspring Tt and tt in 1 : 1.
9	**(d)** 24 cm	Image forms at C = 2f.
10	**(c)** -1 D	P = 1/(-1 m).
11	**(d)** 2×10^{19}	n = 3.2 / 1.6×10^{-19}.
12	**(b)** is reversed	The field direction reverses.
13	**(b)** small fish	Zooplankton are primary consumers.
14	**(d)** Fe_2O_3	Fe_2O_3 loses oxygen.
15	**(d)** anhydrous sodium carbonate	Loses water of crystallisation.
16	**(c)** They have high melting points	Strong forces between ions.
17	**(b)** $CH_2{=}CH_2$	Dehydration.
18	**(d)** herbivores such as the cow	Cellulose needs long digestion.
19	**(d)** auxin affects stem and root cells in opposite ways	Differential response.
20	**(a)** filament	The anther stalk.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Physical change: no new substance, e.g. melting of ice (1); Chemical change: new substance formed, e.g. rusting of iron (1).	1+1
22	Strong acid ionises completely, e.g. HCl (1); Weak acid ionises partially, e.g. CH_3COOH (1).	1+1
23	Bubbles of hydrogen stick to its surface (1); They make it lighter than water (1).	1+1
24	Needs 4 electrons to complete octet (1); Loss/gain requires too much energy (1).	1+1
25	No light, no photosynthesis (1); Iodine test gives no blue-black colour (1).	1+1
26	Quick response avoids danger (1); Does not wait for brain processing (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Al on LHS 1, RHS 2: put 2 before Al (1); $2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Displacement reaction (1).	1+1+1
28	HCl: vigorous bubbling (1); CH_3COOH: slow bubbles (1); Water: no reaction; HCl is a stronger acid (1).	1+1+1
29	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (1); A thin layer of Al_2O_3 forms on the surface (1); It prevents further reaction (anodising thickens it) (1).	1+1+1
30	Formula 3n + 1 (1); Hexane: 19 (1); Decane: 31 (1).	1+1+1
31	Large flat surface to capture light (1); Stomata for gas exchange (1); Chloroplasts rich in chlorophyll in mesophyll (1).	1+1+1
	OR: Thinking and memory (1); Voluntary actions and speech (1); Senses like sight interpretation (1).	1+1+1
32	Spores are tiny and light, so they are dispersed far (1); Many do not find moisture and food (1); Large numbers ensure survival of some (1).	1+1+1

	OR: F1 Tt; F2 ratio 3 : 1 (1); Tall = 225, dwarf = 75 (1); Genotypes TT 75, Tt 150, tt 75 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(a) Displacement (1); (b) Decomposition (1); (c) Double displacement (1); (d) Combination (1); (e) Displacement (1).	1+1+1+1+1
34	Litmus: HCl turns blue to red (1); NaOH and Na_2CO_3 turn red litmus blue (1); Add acid: Na_2CO_3 effervesces, NaOH does not (1); NaCl has no effect on litmus (1); Lime water confirms CO_2 (1).	1+1+1+1+1
35	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Metal oxides are basic (1); Hydrogen is displaced by reactive metals (1).	1+1+1+1+1
36	Tetravalency: four bonds (1); Catenation: long chains, branches, rings (1); Multiple bonds possible (1); Isomerism (1); Strong C-C bonds (1).	1+1+1+1+1
	OR: Destarch the plant in dark (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Iodine test: covered part no blue-black (1); Conclusion: light is necessary (1).	1+1+1+1+1
37	Receptor in skin (1); Sensory neuron to spinal cord (1); Relay and motor neuron (1); Effector muscle moves the foot (1); Brain informed later (1).	1+1+1+1+1
	OR: Binary fission in Amoeba and multiple fission in Plasmodium (1); Budding in yeast and Hydra (1); Fragmentation in Spirogyra and regeneration in Planaria (1); Spore formation in Rhizopus (1); Complex animals have specialised organs which cannot be rebuilt from fragments (1).	1+1+1+1+1
38	Gametes RY, Ry, rY, ry (1); 16 combinations (1); 9 round yellow (1); 3 round green and 3 wrinkled yellow (1); 1 wrinkled green (1).	1+1+1+1+1
	OR: 1/v = -1/12 + 1/20 = -1/30 (1); v = -30 cm (1); m = -v/u = -1.5 (1); Image size = -4.5 cm (1); Real, inverted, magnified; rays: parallel and through F (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** fertilisation	The sperm type decides.
2	**(c)** 0.5	0.75/1.5 = 0.5.
3	**(c)** a single red patch	Monochromatic light does not disperse.
4	**(c)** 2 ohm	R / n.
5	**(a)** to the south	By Fleming's left-hand rule: thumb points to the south.
6	**(d)** eagle	Top consumer.
7	**(c)** $FeSO_4$ (pale green) is formed	Iron displaces copper.
8	**(c)** neutral	No colour change shows neutral.
9	**(a)** keeps air and water away	Barrier effect.
10	**(d)** forms ionic bonds with chlorine	Carbon forms covalent bonds.
11	**(a)** pulmonary artery	Only artery with deoxygenated blood.
12	**(d)** insulin	Diabetes.
13	**(c)** 23	Gametes have half the number.
14	**(a)** both alleles are recessive	Only tt shows dwarf.
15	**(d)** 40 cm	1/v = 1/20 - 1/40 = 1/40.
16	**(c)** dark	No scattering.
17	**(a)** 1100 W	P = V I.
18	**(d)** increases	Faster change gives a larger emf.
19	**(b)** separating biodegradable from non-biodegradable	For proper treatment.
20	**(a)** 5	$C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Unicellular organisms are a single cell and can split (1); Complex organisms have organ systems and need specialised reproductive organs (1).	1+1
22	Dwarf offspring need a t allele from the tall parent (1); Tall parent is Tt (1).	1+1
23	1/f = 1/v + 1/u = -1/30 - 1/30 (1); f = -15 cm (1).	1+1
24	1/f = -1/40 + 1/25 = 3/200 (1); f = 66.7 cm (1).	1+1
25	n = I / e = 0.8 / 1.6×10^{-19} (1); n = 5×10^{18} (1).	1+1
26	At the point of intersection there would be two directions of the field (1); The field has only one direction at a point (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	1/v = -1/15 + 1/20 = -1/60 (1); v = -60 cm; m = -v/u = -3 (1); Size 18 cm; real, inverted, magnified (1).	1+1+1
28	u = infinity, v = -1.5 m (1); f = -1.5 m (1); P = -0.67 D, concave lens (1).	1+1+1
29	R = 3 + 4 + 2 = 9 ohm (1); I = 9 / 9 = 1 A (1); V = I R = 4 V (1).	1+1+1
30	Galvanometer deflects when the magnet moves in (1); It deflects in the opposite direction when withdrawn (1); The magnetic field through the coil changes, inducing a current (1).	1+1+1
31	10 per cent law (1); Deer: 1200 J (1); Tiger: 120 J (1).	1+1+1
	OR: Reaction is endothermic (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); Quick lime is used in cement/whitewash (1).	1+1+1
32	Electrolysis of brine (1); Anode: Cl_2; cathode: H_2 (1); NaOH remains in solution (1).	1+1+1
	OR: Crushing and grinding (1); Removal of gangue (1); Roasting converts ZnS to ZnO (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(a) Hypermetropia (1); (b) 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D (1); (c) Convex lens (1); Image forms at the near point (1).	1+1+1+1+1
34	(a) Parallel = 2 ohm; total = 6 ohm (1); (b) I = 12 / 6 = 2 A (1); (c) V = 2 x 2 = 4 V (1); (d) I in 6 ohm = 4/6 = 0.67 A (1); I in 3 ohm = 4/3 = 1.33 A (1).	1+1+1+1+1
35	Concentric circles around the wire (1); Doubling the current increases the field (1); Doubling the distance weakens the field (1); Right-hand thumb rule: thumb along current (1); Fingers curl in the direction of the field (1).	1+1+1+1+1
36	Phytoplankton, zooplankton, small fish, large fish, fish-eating bird (1); T1: 100,000 J (1); T2: 10,000 J; T3: 1000 J (1); T4: 100 J; T5: 10 J (1); The top level (T5) has the least energy (1).	1+1+1+1+1
	OR: Thermal: $CaCO_3 \longrightarrow CaO + CO_2$ (1); Electrolytic: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Photolytic: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (1); Photolytic used in photography (1); All need energy, so they are endothermic (1).	1+1+1+1+1
37	$CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Heated at 373 K (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$ (1); Uses: plaster, toys, decoration (1); Stored in moisture-proof container (1).	1+1+1+1+1
	OR: Bridges: painting, galvanisation (1); Aluminium: anodising (1); Ships: sacrificial zinc/magnesium plates (1); Zn, Mg are more reactive than iron (1); They corrode instead of iron (1).	1+1+1+1+1
38	$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (sunlight) (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (Ni) (1); $CH_2{=}CH_2 + Br_2 \longrightarrow CH_2Br{-}CH_2Br$ (1); Bromine water decolourised (1); Addition for unsaturated (1).	1+1+1+1+1
	OR: Herbivores have long intestine (1); Carnivores short intestine (1); Cellulose digestion in herbivores (1); Caecum is large in herbivores (1); Carnivores digest meat easily (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** pH 1	Lower pH means more H^+.
2	**(b)** sodium	Only highly reactive metals react with cold water.
3	**(a)** 10	3 x 3 + 1 = 10.
4	**(a)** oxygen	Oxygen is a product.
5	**(d)** relay neuron	Relay neuron.
6	**(a)** Plasmodium	Multiple fission under unfavourable conditions.
7	**(a)** both Tt	3 : 1 ratio arises from Tt x Tt.
8	**(d)** 30 cm	Same size at C.
9	**(c)** +2 D	1/f = 1/25 - 1/50 = 1/50 gives +2 D.
10	**(a)** 2 A	I = 600 / 300.
11	**(c)** larger than B	Nearer, stronger.
12	**(a)** Plant, caterpillar, bird, hawk	Order of feeding.
13	**(c)** C	Carbon takes oxygen from ZnO.
14	**(b)** thermal decomposition	Heat decomposes baking soda.
15	**(a)** +2	It loses two electrons.
16	**(b)** CH_3COOH	Oxidation.
17	**(b)** cytoplasm	Cytoplasm.
18	**(b)** does not occur	Auxin source is lost.
19	**(c)** unisexual (male)	Only male parts are present.
20	**(a)** 1/2	Each birth is independent.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Grass, grasshopper, frog, snake (1); Tertiary consumer: snake (1).	1+1
22	$2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Hydrogen gas (1).	1+1
23	Adds quick lime or slaked lime (1); Basic substances neutralise the acid (1).	1+1
24	Ions are free in solution (1); Fixed in the solid lattice (1).	1+1
25	Add bromine water (1); Unsaturated decolourises it (1).	1+1
26	Cellulose digestion takes longer (1); Meat is easier to digest (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Soap: fatty acid salts; detergent: sulphonate salts (1); Soap forms scum in hard water (1); Detergents don't (1).	1+1+1
28	Mixing reduces the oxygen supplied to the body (1); Birds and mammals need more energy (1); Four chambers prevent mixing (1).	1+1+1
29	Low thyroxine causes pituitary to release more TSH (1); Thyroid makes thyroxine (1); Level returns to normal (1).	1+1+1
30	Male gametes are produced in large numbers (1); They are small and motile and many are lost before reaching the egg (1); Female gametes are few, larger and rich in food (1).	1+1+1
31	Round yellow 9/16 = 450 (1); Round green 3/16 = 150, wrinkled yellow = 150 (1); Wrinkled green 1/16 = 50 (1).	1+1+1
	OR: 1/v = -1/10 - 1/20 = -3/20 (1); v = -6.67 cm (1); m = v/u = 1/3; virtual, erect, diminished (1).	1+1+1
32	At noon light has a short path in the atmosphere (1); At sunset it travels a long path (1); Most blue is scattered away, so red remains (1).	1+1+1
	OR: R = rho L / A (1); Greater area gives lower resistance (1); Less heat I^2 R t, so the wire does not overheat (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Sugar rises after meal (1); Insulin released (1); Sugar stored/used (1); Diabetes: not enough insulin (1); Insulin injections treat (1).	1+1+1+1+1
34	Ovaries produce eggs and hormones (1); Oviduct is the site of fertilisation (1); Uterus receives and nourishes the embryo (1); Placenta exchanges materials with mother's blood (1); Cervix and vagina form the birth canal (1).	1+1+1+1+1
35	Genotypes TT, Tt and tt (1); Phenotypes tall and dwarf (1); T is dominant (1); t is recessive (1); T and t are alleles (1).	1+1+1+1+1
36	(a) Water, kerosene, glass, diamond (1); (b) v = 3×10^8/2.42 (1); v = 1.24×10^8 m/s (1); (c) Diamond (1); Greater n means greater bending towards the normal (1).	1+1+1+1+1
	OR: u = infinity, v = -80 cm (1); f = -80 cm (1); P = -1.25 D (1); Near point is 25 cm, normal (1); No lens needed for reading, but a concave lens for distance (1).	1+1+1+1+1
37	H = I^2 R t (1); H = 4 x 50 x 600 (1); H = 1.2×10^5 J (1); In calories: 1.2×10^5 / 4.2 (1); = 28571 cal (1).	1+1+1+1+1
	OR: Mains: live, neutral and earth wires (1); Appliances are connected in parallel (1); Geyser current = 2000 / 220 = 9.1 A (1); More than 5 A, so needs the 15 A line (1); Fuse in the live wire melts for excess current (1).	1+1+1+1+1
38	Sun gives energy to producers (1); Producers convert it into chemical energy (1); Passed on to consumers (1); Energy lost as heat at each step (1); Heat energy cannot be used by organisms, so it is not recycled (1).	1+1+1+1+1
	OR: (i) Oxidised: H_2; reduced: CuO (1); (i) Oxidising agent CuO; reducing agent H_2 (1); (ii) Oxidised: CO (1); (ii) Reduced: Fe_2O_3 (1); (ii) Oxidising agent Fe_2O_3; reducing agent CO (1).	1+1+1+1+1

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